MODULE 2 SPINE AND THORACIC CAGE

The Vertebral Column and Thoracic Cage.

AXIAL SKELETON

The vertebral column is the central axis of the body, and the thoracic cage is built onto it. This chapter covers the regions and curvatures, a typical vertebra, each region of vertebrae in detail, and the sternum and ribs.

BY THE END

1. Name the five regions of the vertebral column and the number of vertebrae in each.
2. Identify the parts of a typical vertebra and the structures between adjacent vertebrae.
3. Tell a cervical, thoracic, and lumbar vertebra apart from a single feature.
4. Name the four curvatures and sort them as primary or secondary.
5. Identify the parts of the sternum and classify the ribs by anterior attachment.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Axial Skeleton, Spine

drsrennie-stack.github.io/new-build-bio4-solano/axial-concept-videos.html

The Vertebral Column, an Overview

Early in development the column has 33 vertebrae. In the adult, 5 sacral and 4 coccygeal vertebrae fuse, leaving 26 bones: 24 separate vertebrae in three regions, plus the sacrum and the coccyx.

THE FIVE REGIONS, SUPERIOR TO INFERIOR

Region	Vertebrae	Description
Cervical region	7, C1 to C7	The neck
Thoracic region	12, T1 to T12	The chest. Each vertebra articulates with a pair of ribs
Lumbar region	5, L1 to L5	The lower back
Sacrum	5, fused	A single triangular bone wedged between the hip bones
Coccyx	4, fused	The tailbone

Intervertebral discs are fibrocartilage pads between adjacent vertebral bodies.

The Curvatures

The adult spine has four front-to-back curves that act like a spring, giving the column strength, flexibility, and balance. Primary curvatures are present at birth. Secondary curvatures develop with the milestones of infancy.

TYPE AND TIMING

Curvature	Type	When it develops
Cervical curvature	Secondary	Develops as an infant holds up its head
Thoracic curvature	Primary	Present from birth
Lumbar curvature	Secondary	Develops as a child stands and walks
Sacral curvature	Primary	Present from birth

KEY POINT

The primary curves are the two you were born with, and they both curve the same way as the fetal C shape. The secondary curves are the two you earned, one by lifting your head and one by standing up.

A Typical Vertebra

Parts of a vertebra

FROM THE BODY OUTWARD

Part	What it is
Body	The thick, weight-bearing anterior portion. Also called the centrum
Vertebral arch	The bony ring projecting from the back of the body
Pedicles	The two short bars connecting the arch to the body
Laminae	The two flat plates that complete the arch behind
Vertebral foramen	The opening enclosed by the body and arch. The spinal cord runs through it
Spinous process	The single projection pointing backward from the arch
Transverse processes	The two projections pointing laterally from the arch
Articular processes	The superior and inferior pairs that join one vertebra to the next

Between the vertebrae

THE SPACES AND THE DISCS

Structure	What it is
Vertebral canal	The stacked vertebral foramina, the channel that holds the spinal cord
Intervertebral foramina	The gaps between stacked vertebrae where the spinal nerves exit
Intervertebral disc	A fibrocartilage pad between two vertebral bodies, a shock absorber
Anulus fibrosus	The tough outer ring of the disc
Nucleus pulposus	The soft, gel-like core of the disc

Regional Vertebrae

Each region has a vertebra shaped for its job. One feature is enough to place a loose vertebra.

BODY, SPINOUS PROCESS, AND THE GIVEAWAY FEATURE

Region	Body	Spinous process	Distinctive feature
Cervical, C1 to C7	Small and oval, the smallest vertebrae	Short, often bifid or forked	Transverse foramina in the transverse processes, which pass the vertebral arteries
Thoracic, T1 to T12	Heart-shaped	Long, angled sharply downward	Costal facets on the body and transverse processes where the ribs articulate
Lumbar, L1 to L5	Thick and kidney-shaped, the largest vertebrae	Short and blunt	Massive build for weight bearing. No costal facets and no transverse foramina

The specialized cervical vertebrae

THE TWO THAT ARE NOT TYPICAL

Structure	What it is
Atlas (C1)	A ring with no body and no spinous process. It holds up the skull
Axis (C2)	Bears the dens, an upward bony peg
Dens	The odontoid process, the peg of the axis that the atlas pivots around for the no rotation
Atlanto-occipital joint	Where the atlas meets the occipital condyles. The joint for the yes nod
Atlanto-axial joint	Where the atlas pivots on the dens of the axis. The joint for the no head rotation
Vertebra prominens (C7)	A long spinous process you can feel at the base of the neck

Sacrum and coccyx

THE FUSED REGIONS

Structure	What it is
Sacrum	Five fused vertebrae forming a triangular bone wedged between the hip bones
Sacral promontory	The anterior ridge of the first sacral vertebra
Median sacral crest	The fused spinous processes along the posterior surface
Sacral foramina	The openings that pass the sacral nerves
Sacral canal	The continuation of the vertebral canal through the sacrum
Sacral hiatus	The opening at the inferior end of the sacral canal
Auricular surface	The ear-shaped surface that joins the hip bone at the sacroiliac joint
Coccyx	The fused tailbone, an attachment point for pelvic ligaments and muscles

The Thoracic Cage

The thoracic cage is the sternum in front, the ribs on the sides, and the thoracic vertebrae behind. It shields the heart and lungs.

The sternum

THREE FUSED PARTS AND THEIR LANDMARKS

Part	What it is
Sternum	The breastbone, a flat bone of three fused parts
Manubrium	The superior part. Articulates with the clavicles and the first ribs
Body	The long middle part of the sternum
Xiphoid process	The small inferior tip, a muscle attachment. Cartilage that ossifies with age
Jugular notch	The central dip at the top of the manubrium, easily felt
Sternal angle	The ridge where the manubrium meets the body. A landmark level with the second rib

The ribs

PARTS OF A RIB

Part	What it is
Ribs	Twelve pairs of flat, curved bones forming the walls of the cage
Head	The posterior end of a rib. Articulates with the vertebral bodies
Tubercle	The knob near the head. Articulates with a transverse process
Costal cartilage	The hyaline cartilage that connects a rib toward the sternum

THE THREE GROUPS, BY ANTERIOR ATTACHMENT

Rib group	Pairs	Anterior attachment
True ribs	1 to 7	Attach to the sternum directly, each by its own costal cartilage
False ribs	8 to 12	Do not reach the sternum directly. Pairs 8 to 10 join the costal cartilage of the rib above
Floating ribs	11 and 12	No anterior attachment at all. A subset of the false ribs

THE SPINE PROTECTS A CORD, THE CAGE PROTECTS A PUMP

The vertebral canal carries the spinal cord, so a vertebral fracture is dangerous for what runs inside it, not only for the bone. A herniated disc occurs when the soft nucleus pulposus pushes through a torn anulus fibrosus and presses on a spinal nerve at an intervertebral foramen, which is why the pain travels down a limb. The dens of the axis is held against the atlas by a ligament, and a sharp blow can fracture it, a serious injury so close to the brainstem. The sternal angle is a landmark a clinician feels for, sitting at the level of the second rib, the starting point for counting ribs, and the xiphoid process guides hand

position for chest compressions in CPR. Abnormal curvatures have names: scoliosis is a sideways curve, kyphosis is an exaggerated thoracic curve, and lordosis is an exaggerated lumbar curve.